

# Activity – 13

1. Writing a menu driven c program to count the even digits , odd digits and sum of even digits and odd digits.

```
#include <stdio.h>
int main(){
    int num, choice;
    printf("\n Look here first \n");
    printf("\nEnter the number.\n");
    scanf("%d", &num);

    do{
        printf("\n -----Menu----- \n");
        printf("\n Enter 1 for counting even and odd numbers. \n");
        printf("\n Enter 2 for getting the sum of even and odd digits. \n");
        scanf("%d", &choice);

        switch(choice){
            case 1: {
                int temp = num, digit, evencount=0, oddcount=0;
                while (temp > 0){
                    digit = temp % 10;
                    if(digit % 2 == 0) evencount++;
                    else oddcount++;
                    temp /= 10;
                }
                printf("Evencount: %d, oddcount: %d\n", evencount, oddcount);
                break;
            }
            case 2: {
                int temp = num, digit, evensum=0, oddsum=0;
                while (temp > 0){
                    digit = temp % 10;
                    if(digit % 2 == 0) evensum += digit;
                    else oddsum += digit;
                    temp /= 10;
                }
                printf("Evensum: %d, oddsum: %d\n", evensum, oddsum);
                break;
            }
            case 3:
                printf("Exiting the program.\n");
                break;
            default:
                printf("Invalid choice.\n");
                break;
        }
    }while(choice != 3);
    return 0;
}
```

Compilation and execution

```
(kali㉿kali)-[~]
$ vi newd5.c

(kali㉿kali)-[~]
$ gcc newd5.c -o newd5

(kali㉿kali)-[~]
$ ./newd5

Look here first

Enter the number.
123456

——Menu——

Enter 1 for counting even and odd numbers.

Enter 2 for getting the sum of even and odd digits.
2
Evensum: 12, oddsum: 9

——Menu——

Enter 1 for counting even and odd numbers.

Enter 2 for getting the sum of even and odd digits.
1
Evencount: 3, oddcount: 3

——Menu——

Enter 1 for counting even and odd numbers.

Enter 2 for getting the sum of even and odd digits.
3
Exiting the program.
```

## Test case table

| INPUT | Output(expected) | Output           | Result |
|-------|------------------|------------------|--------|
| 14567 | (2, 3), (10, 13) | (2, 3), (10, 13) | PASS   |

|      |                 |                    |      |
|------|-----------------|--------------------|------|
| 145  | (1, 2), (4, 6)  | (1, 2),<br>(4, 6)  | PASS |
| 1897 | (1, 3), (8, 17) | (1, 3),<br>(8, 17) | PASS |

## Algorithm

1.START

2.Input number , num

3.show menu(count digits or sum digits or exit)

4.read choice

5.if choice = 1, count even and odd digits.

6. if choice = 2, sum up even digits and sum up odd digits.

7. if choice = 3, exit

8. STOP